Sorrentino Alessio

MSC STUDENT IN AUTOMATION AND INTELLINGENT CYBER-PHYSICAL SYSTEMS AT POLITECNICO DI TORINO

CONTACT

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Torino & Rome

LANGUAGES

Italian	Native language
English	B2 - Ielts Certificate

Engineer, problem solver and dreamer just enough. After graduating from the Polytechnic of Turin with honors, I chose to specialize in Automation and Cyber-Physical Systems because I believe that the future lies in the perfect integration between hardware and software. My approach? Looking beyond the screen: whether it's optimizing a smart home or designing complex systems, my goal is always efficiency and reliability. Point at the top — realities such as NASA and ESA are my reference for quality standards — but I am always open to new technological challenges that require analytical rigor and a spirit of initiative.

CERTIFICATION

2024 | IELTS - ENGLISH CERTIFICATION B2

Intermediate-higher knowledge of the English language, corresponding to level B2 of the Common European Framework of Reference for Languages (CEFR)

2021 - 2022 | CISCO CERTIFICATION: IT ESSENTIAL

Technical skills in the assembly, maintenance and updating of PCs, installation and management of operating systems (Windows, Mac, Linux) and configuration of networks, mobile devices and printers. Experience in the safe use of tools, advanced troubleshooting and implementation of IT security measures.

PROFESSIONAL SKILLS

- Programming: C, Java, Python, Bash, SQL, RISC-V & MIPS Assembly, Verilog.
- Embedded & Hardware: ARM Cortex-M3, Arduino, Real-time control, ADC, I/O Interfacing.
- Web & Data: RESTful API Design, JSON, MongoDB (MQL), HTML, Streamlit.
- IoT & Networking: MQTT Protocol, Wireshark, Network Troubleshooting.
- Scientific Computing: MATLAB, Simulink, Digital Signal Processing (DSP).
- Tools: Git, Keil IDE, JPA/Hibernate, H2 Database.

EDUCATION

2025 - | POLITECNICO DI TORINO
Student in Automation and Intellingent Cyber-Physical Systems

2022 - 2025 | POLITECNICO DI TORINO
Degree with honors in Computer Engineering

2017 - 2022 | LICEO LEONARDO DA VINCI - TERRACINA
Scientific High School Diploma

TECHNICAL PROJECTS

IoT & Smart Home Automation | Polytechnic of Turin :
Engineered a distributed home automation system using MQTT/HTTP protocols.
- Developed a scalable CherryPy backend and integrated Arduino nodes for environmental monitoring and autonomous contextual automation.

Embedded Systems: Real-Time Tetris | Polytechnic of Turin :
Developed a real-time Tetris game on ARM Cortex-M3 (LPC1768) in C.
- Implemented ADC-based speed control and hardware audio playback, optimizing performance for resource-constrained environments.

Advanced Simulator Engine (Java & JPA) | Polytechnic of Turin :
Designed a high-performance analytical engine for a Game of Life simulator using Java Stream API.
- Managed the persistence layer via JPA/Hibernate and H2 database to track complex entity states and time series.

Spectral Analysis & Digital Filter Design | Polytechnic of Turin :
Implemented FIR digital filters and FFT algorithms in MATLAB for audio signal processing.
- Validated frequency responses using Gaussian noise and real audio tracks to ensure attenuation accuracy.

PROFESSIONAL EXPERIENCE

2025 | Computer Science Teaching Assistant | Polytechnic of Turin :
Supported students during laboratory experiments and exercise sessions.
- Provided technical assistance during examinations and managed grading/reception hours.

2022 – 2025 | Laboratory Activities | Polytechnic of Turin :
Collaborated in multi-disciplinary teams to complete technical challenges and practical simulations related to the engineering curriculum.

2017 – 2022 | Seasonal Worker | Various Establishments :
Developed early professional skills, focusing on teamwork and client communication in high-pressure summer environments.